

Shrinkage Estimation of Covariance Matrices: Economic Loss Minimization with Reinforcement Learning

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Feature-conditional covariance estimation

- ▶ **Covariance matrices** are fundamental for portfolio optimization, return prediction, and risk management.
- ▶ Existing methods use **only** the sample of **asset returns** for covariance estimation.
- ▶ Considering **additional macro features** has the potential to **improve** the estimator.
- ▶ **Idea:** Fit the **feature-conditional** estimator with respect to the investor's **economic objective**, not the statistical properties of the matrix.

Main Result

Our approach **closes up to 32% of the gap** between the best existing data-driven method and the oracle estimator and attains the lowest realized volatility for every universe size.

Covariance matrix shrinkage

- ▶ Sample covariance with K assets and T days: $K(K + 1)/2$ free parameters — noisy when $K \asymp T$.
- ▶ To improve the OOS performance, investors use covariance matrix shrinkage estimation — an approach of **modifying** the **sample covariance matrix**, using an **economic prior**.
- ▶ An example of a prior is assuming a **zero correlation across all assets** and each asset's **variance** being equal to the **average sample variance** across the whole universe.
- ▶ For shrinkage intensity $a_t \in [0, 1]$, the shrunk covariance matrix is:

$$\hat{\Sigma}_t(a_t) = (1 - a_t) S_t + a_t \bar{s}_t^2 I_K, \quad \bar{s}_t^2 = \frac{1}{K} \text{tr}(S_t),$$

where $S_t \in \mathbb{R}^{K \times K}$ — sample covariance from the last T days of returns; I_K — identity.

- ▶ $a_t = 0$: pure sample covariance — **low bias, high variance**.
- ▶ $a_t = 1$: fully shrunk — **high bias, low variance**.

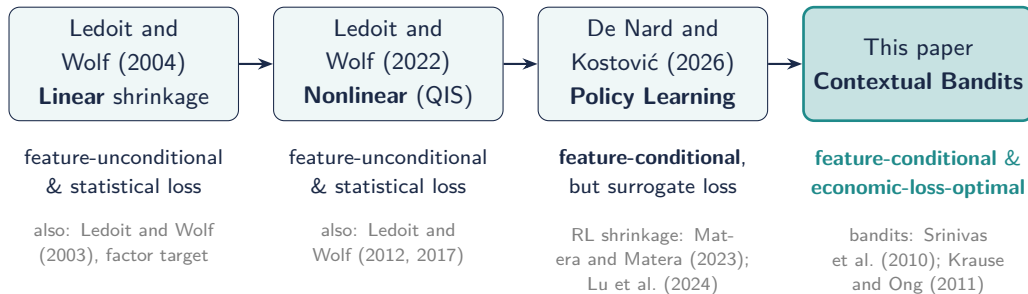
Another view on shrinkage

- ▶ We view “shrinkage” as **feature-conditional learning** **modifying the sample covariance** to improve portfolio performance.
- ▶ Learning all $K(K + 1)/2$ entries of the covariance matrix using additional features would **result in overfitting**.
- ▶ We can shift **all entries** in **one learned direction**: a Bayesian prior on the covariance, whose weight a_t is learned from the current market state.
- ▶ Thereby, **one parameter controls** all $K(K - 1)/2$ correlations and K variances.

Main Question

How should an investor choose a_t , so that the portfolio built from $\hat{\Sigma}_t(a_t)$ attains the lowest risk?

Where we stand in the literature



Economic idea: Optimal shrinkage is feature-conditional

- ▶ After a **market shock**, correlations are usually **elevated** — a stylized fact of financial markets.
- ▶ In this period, one should **reduce** the **estimated correlations**, shrinking more severely.
- ▶ In **calm markets** the sample structure is informative — shrink less.
- ▶ Thus, the optimal a_t^* **moves with the economy**:

$$a_t^* = a^*(s_t), \quad s_t \text{ — current market state}$$

- ▶ Optimal shrinkage $\Rightarrow$ **feature-dependent**.

Same formula, two regimes

$$\hat{\sigma}_{ij,t}(a_t) = (1 - a_t) s_{ij,t}$$

- ▶ Post-crisis: $s_{ij,t}$ inflated $\Rightarrow$ large a_t .
- ▶ Calm: $s_{ij,t}$ trustworthy $\Rightarrow$ small a_t .

Cross-validation / Ledoit–Wolf pick *one* a for both.

Problem setup: Economic objective

- Choose shrinkage intensity a_t to minimize the realized out-of-sample portfolio risk:

$$\hat{w}_t(a_t) := w(\hat{\Sigma}_t(a_t)), \quad \min_{a_t \in \mathcal{A}} \hat{w}_t(a_t)^\top \Sigma_t^* \hat{w}_t(a_t)$$

Σ_t^* — true (unobservable) covariance of the data-generating process; $w(\cdot)$ — minimum-variance optimizer (GMV or long-only); $\mathcal{A}$ — admissible intensities.

- Oracle portfolio $w_t^* := w(\Sigma_t^*)$ — best achievable.

Regret

$$\mathcal{R}_t(a_t) = \sqrt{\hat{w}_t(a_t)^\top \Sigma_t^* \hat{w}_t(a_t)} - \sqrt{(w_t^*)^\top \Sigma_t^* w_t^*} \geq 0$$

- Supervised label a_t^* is **not observable** from the data directly.
- We only ever observe the *consequences* of a decision, never the correct decision to take.

Problem setup: Reward proxy

Ex post, every a_t can be scored

1. Choose a shrinkage intensity a_t .
2. Build $\hat{\Sigma}_t(a_t)$.
3. Construct the portfolio $\hat{w}_t(a_t)$.
4. Backtest $L = 21$ days $\rightarrow$ realized volatility:

$$r(s_t, a_t) = -\sigma(\hat{w}_t(a_t)^\top R_{t+1:t+L})$$

s_t — market state; $R_{t+1:t+L}$ — realized asset returns over the holding period; $\sigma(\cdot)$ — sample portfolio volatility.

Why is this expected to approximate the *true* reward?

- ▶ Σ_t^* **changes slowly** $\Rightarrow$ realized variance $\approx \hat{w}_t(a_t)^\top \Sigma_t^* \hat{w}_t(a_t)$.
- ▶ Only the *ranking* of intensities matters, not the level of volatility.
- ▶ Backtesting is free $\Rightarrow$ the whole **reward curve** $a_t \mapsto r(s_t, a_t)$ is observable for every past period.

Why not cross-validation?

- ▶ Cross-validation assumes that one **unconditional shrinkage intensity** is optimal across all folds.
- ▶ It is a model-selection criterion based on historical performance — it **averages** over the very **regimes we wish to distinguish**.
- ▶ Same critique applies to any feature-unconditional estimator, including Ledoit–Wolf.
- ▶ However, a_t^* is **driven by the market state** $s_t \implies$ learn a forward-looking map from **currently observed features** O_t (e.g. lagged reward curves, VIX, inflation) to $\hat{a}_t$.

Learning Environment:

- ▶ In each step, we transition to a new state $s_t \sim p(\cdot \mid O_t, U_t)$.
- ▶ We treat absence of U_t as if we observed a noisy version of the state $O_t \sim \mu(\cdot \mid s_t)$.
- ▶ Policy acts greedily w.r.t. the **learned reward curve**:

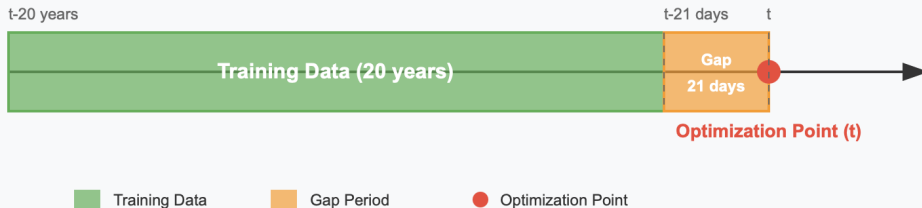
$$\hat{a}_t = \pi(O_t) \in \arg \max_{a_t \in \mathcal{A}} \hat{r}(O_t, a_t).$$

- ▶ **No exploration needed** — full offline coverage, no evaluation costs.

Delayed feedback

- **Delayed feedback** — last reward curve is lagged by $L = 21$ steps:

Time Series Split Configuration



Why RL → Contextual Bandits for this task?

- ▶ In absence of costs, actions **do not affect future portfolios**:

$$P(s_{t+1} \mid s_t, a_t) = P(s_{t+1} \mid s_t).$$

- ▶ Bellman optimality then collapses to a **single step**:

$$a_t^* = \pi^*(s_t) \in \arg \max_{a_t} \left[r(s_t, a_t) + \underbrace{\gamma \sum_{s_{t+1}} P(s_{t+1} \mid s_t) V^*(s_{t+1})}_{\text{independent of } a_t} \right] = \arg \max_{a_t} r(s_t, a_t)$$

π^* — optimal policy; V^* — optimal value function; γ — discount factor.

- ▶ Sequential RL → **Contextual Bandits** (one-step).
- ▶ **Today's shrinkage** does not depend on **previously chosen shrinkage** — each rebalancing is its own decision.

Why not Policy Learning? (De Nard and Kostović, 2026)

- PL: use supervised learning to fit $\mathcal{L}(a_t^*, g(O_t))$ with MSE.

If discretization error $\equiv 0$, the regret splits as

$$C(\hat{\pi}_m) - C(\pi_{\text{oracle}}) = \underbrace{C(\pi_m^*) - C(\pi_{\text{oracle}})}_{\text{risk mismatch}} + \underbrace{C(\hat{\pi}_m) - C(\pi_m^*)}_{\text{estimation error}}$$

$C(\cdot)$ — expected cost; π_m^* — population-optimal policy of method m ; $\hat{\pi}_m$ — its finite-sample estimate.

Policy Learning

Risk mismatch > 0

Survives even as $n \rightarrow \infty$.

Contextual Bandits

Risk mismatch $= 0$

For any sample size n .

- Under **full observability**, the methods are expected to produce the **same result**.
- Our learning task is in **partially observable** and **low signal-to-noise** environment:
 $O_t \sim \mu(\cdot \mid s_t)$.
- The estimator is **Bayes-optimal** w.r.t. the observed features O_t — proof in the paper.

Estimation pipeline

1. **Collect reward curves** — backtest every shrinkage intensity across a discretized grid (e.g., 100 actions) on every past day.
2. **Encode each curve by Random Fourier Features** — the coefficients concatenated with additional features (e.g., VIX) are the **latent state of the market**.
3. **Fit a VARX, roll it forward $L = 21$ steps** — forecasts under the delayed-feedback gap.
4. **Select the shrinkage intensity** — maximizer of the predicted reward curve:

$$\hat{a}_t \in \arg \max_{a_t \in \mathcal{A}_d} \hat{Y}_{t,a_t},$$

$\hat{Y}_{t,a_t}$ — predicted reward of intensity a_t ; $\mathcal{A}_d$ — discretized grid.

- Detailed algorithm in the appendix — happy to explain in detail.

Empirical study

Empirical study: Setup

Data

- ▶ CRSP daily US stock returns, $K \in \{100, 500, 1000, 2000\}$ largest stocks, universe rebuilt at each rebalancing.
- ▶ Train: 1981–1999.
Test: 2000–2024.
- ▶ Monthly rebalancing ($L = 21\text{d}$) of Global & Long-only MinVar.

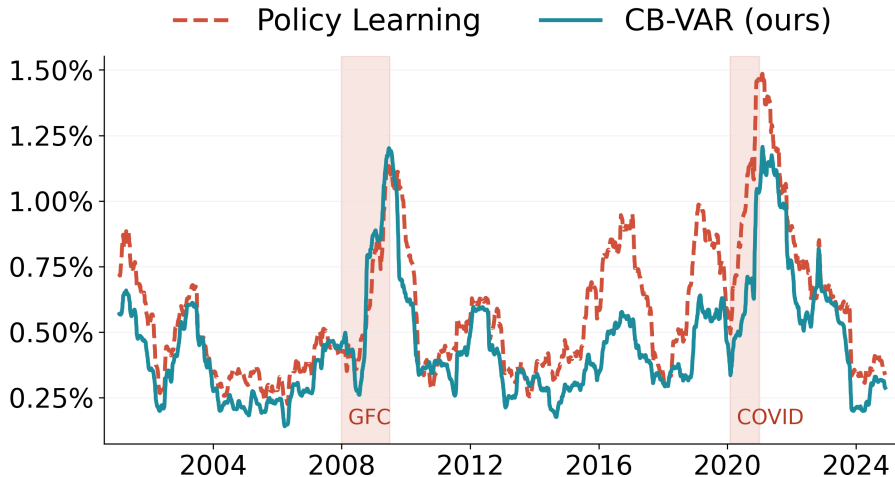
Benchmarks

- ▶ LW — linear shrinkage (Ledoit and Wolf, 2004).
- ▶ CV — cross-validated intensity.
- ▶ PL — policy learning (De Nard and Kostović, 2026).
- ▶ CB-VAR — **ours**.
- ▶ Oracle — knows a_t^* (ex-post best a_t *within the shrinkage family*).

Metrics

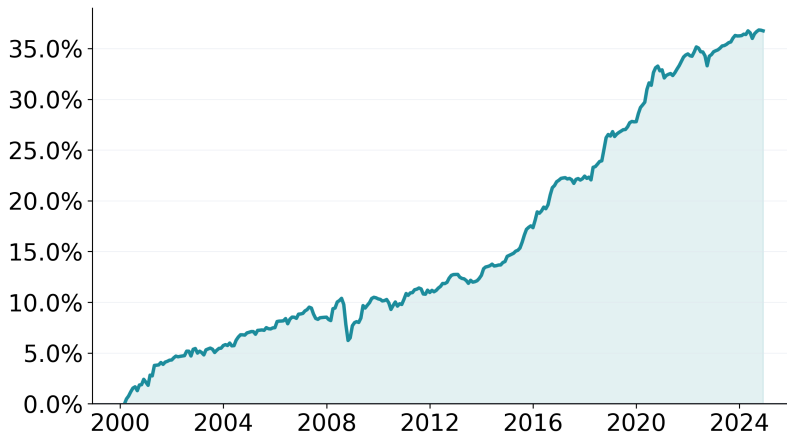
- ▶ Realized annualized volatility.
- ▶ Gap closed = share of the PL-to-oracle gap removed by CB-VAR.
- ▶ 99% block-bootstrap CI (42-day blocks).

Regret vs oracle over time ($K = 500$, 1Y moving average)



- CB-VAR sits below PL and the spread widens exactly in the GFC and COVID regimes.

Cumulative regret saved vs policy learning ($K = 500$)



- Regret is **additive across rebalancings**: small difference in volatility per month compounds into a large cumulative improvement.

Key result: Realized annualized volatility, GMV (%)

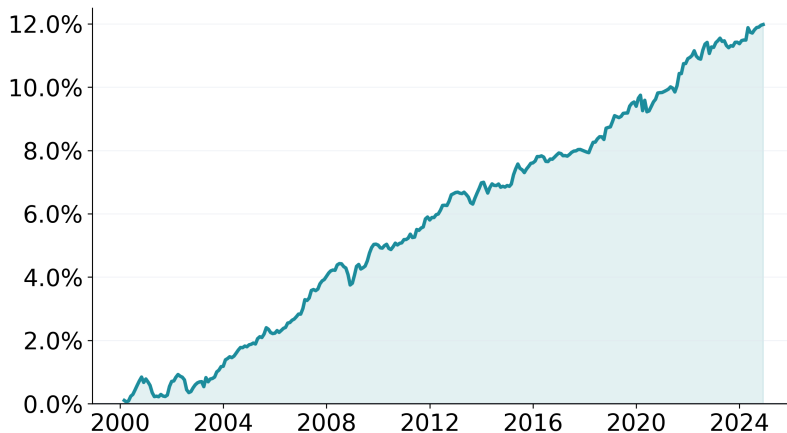
K	LW	CV	PL	CB-VAR	Oracle	Gap closed
100	13.37	12.85	12.82	12.55	11.82	27.31%
500	11.69	11.31	11.21	10.97	10.46	31.69%
1000	10.16	10.16	10.20	10.07	9.77	30.03%
2000	10.07	10.15	10.16	10.06	9.78	26.85%

99% bootstrap CIs exclude 0 for all K : $K = 500$ CI is $[18.0\%; 43.2\%]$.

Result

Lowest volatility among all benchmark estimators, for every universe size.

Nonlinear shrinkage (QIS): Same picture ($K = 500$)



► Gap closed vs PL: 9.0% / **39.6%** / 26.2% / 25.8% across K .

Long-only minimum variance, linear shrinkage (%)

K	LW	CV	PL	CB-VAR	Oracle	Gap closed
100	13.75	13.76	13.62	13.45	13.02	27.93%
500	14.71	13.65	13.00	12.98	12.09	3.05%
1000	13.92	14.37	12.62	12.53	11.42	7.01%

- Still the **lowest realized volatility**, but the edge over PL narrows.
- Intuition: the long-only constraint already acts as a shrinkage (Jagannathan and Ma, 2003), so there is **less left for a_t to shift**.

Three points to take away

- ▶ **Covariance estimation is a decision problem.** The estimator should be selected by the **investor's economic objective**, not a statistical fit.
- ▶ **Returns are not the only input.** Observable market features identify the current state, and the **optimal shrinkage intensity is a function of that state** — any single global intensity is dominated.
- ▶ **The resulting problem is a Contextual Bandit without exploration.** Every intensity can be evaluated offline. Acting greedily on the **learned reward curve** is Bayes-optimal with respect to the observed features.

Result

CB-VAR **closes up to 32% of the gap** between policy learning and the oracle and attains the lowest realized volatility for every universe size.

Thank you for your attention!

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Happy to discuss our work in detail.

Appendix

Appendix A — Pipeline, steps 1–2

Estimate $\hat{a}_t$ from the dataset $(X_u, R_u)_{t-H \leq u < t}$, $R_u \in \mathbb{R}^K$

1 · Collect reward curves $Y_{u,a}$

$$\hat{\Sigma} = \Sigma(a, R_{u-T+1:u}), \quad w(\hat{\Sigma}) \in \mathbb{R}^K, \quad Y_{u,a} := r(s_u, a) = \text{Backtest}(w(\hat{\Sigma}), R_{u+1:u+L})$$

H — training history; T — estimation window (sample size); L — holding period; R — returns of the K stocks. Full offline coverage / no evaluation cost $\Rightarrow$ no exploration needed.

2 · Encode the curve by RFF, extract the latent state

$$Y_{u,a} \sim \mathcal{GP}(\mu, k) \xrightarrow{\text{RFF}} Y_{u,a} \overset{\text{Ridge}}{\approx} \phi(a)^\top \hat{\beta}_u, \quad \Phi = [\phi(a)]_{a \in \mathcal{A}_d} \in \mathbb{R}^{|\mathcal{A}_d| \times p} \quad (\text{same for every } u)$$

$$\text{Latent state:} \quad \Psi_u := [\hat{\beta}_u^\top \ X_u] \in \mathbb{R}^{p+d}$$

$\mathcal{A}_d$ — discretized action grid; $\phi(a)$ — random Fourier features of the action; $\hat{\beta}_u$ — ridge coefficients; $X_u \in \mathbb{R}^d$ — external features. Fixed dictionary $\Rightarrow$ only $\hat{\beta}_u$ varies over regimes.

Appendix A — Pipeline, steps 3–4

3 · Fit VARX and roll forward ($L = 21$)

$f_\theta = \text{VARX}(1)$ propagates the latent state through L steps, with X as the exogenous block. For $i = 1, \dots, L$:

$$\hat{\beta}_{t-L+i} = f_\theta(\Psi_{t-L+i-1}^\top), \quad \hat{\Psi}_{t-L+i} = [\hat{\beta}_{t-L+i}^\top \ X_{t-L+i}]$$

θ — VARX coefficients, ridge-penalized. Autoregressive regimes & L -step delay of the last reward curve. Latent-state half-life at $K = 500$: 59% (CB-VAR) vs 3% (PL).

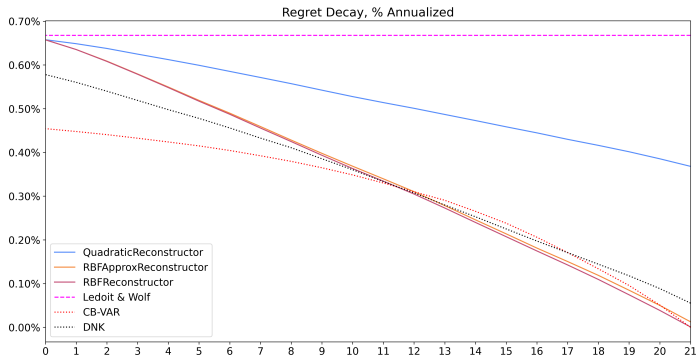
4 · Select the shrinkage a^*

$$\hat{\beta}_t \in \mathbb{R}^p \implies \hat{Y}_t = \Phi \hat{\beta}_t \in \mathbb{R}^{|\mathcal{A}_d|}, \quad \hat{a}_t = \arg \max_{a_t \in \mathcal{A}_d} \hat{Y}_{t,a_t}$$

Argmax on the learned reward $\rightarrow$ greedy w.r.t. $Q^*(s_t, a_t) \implies$ Bayes-optimal.

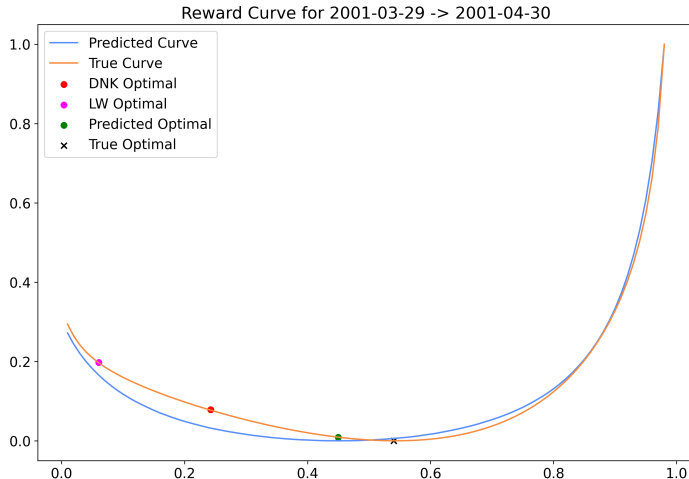
- Grids: $\mathcal{A}_d^L = \{0.00, 0.01, \dots, 1.00\}$ (linear), $\mathcal{A}_d^{NL} = \{0.5, 0.55, \dots, 2.0\}$ (nonlinear); 10% of the borders trimmed for stability.
- Why RFF and not a full GP: negligible reconstruction error (Appendix C), but allows for finite-dimensional state for the VARX.

Appendix B — Where does the edge come from? Regret vs information leakage ($K = 500$)



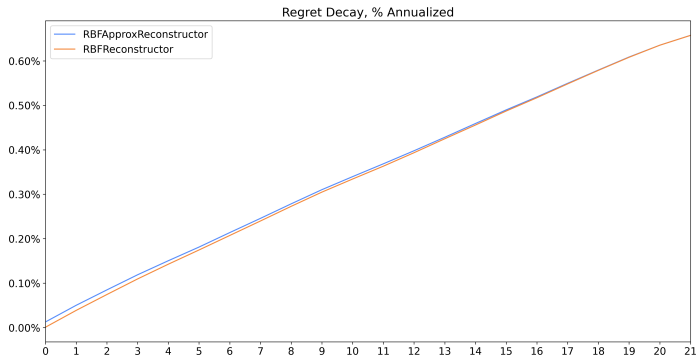
- Point 0: no lookahead; point 21: the full future 21 days leaked into the model.
- With little information, modelling the *curve* beats forecasting the *optimum* — the advantage of framing learning as a decision task.

Appendix C — Example of a reward curve (linear shrinkage)



- Empirically convex in a_t (see Appendix H); only the position of the minimum matters for the decision.

Appendix C — GP with RBF kernel vs RFF-Ridge approximation



- Reconstruction difference between the full Gaussian process and the RFF-Ridge pipeline is negligible.

Appendix D — Linear shrinkage, GMV: full tables

Realized annualized volatility (%)

K	LW	LWCV	PL	CB-VAR	Oracle	Ridgeless
100	13.37	12.85	12.82	12.55	11.82	14.82
500	11.69	11.31	11.21	10.97	10.46	13.49
1000	10.16	10.16	10.20	10.07	9.77	11.99
2000	10.07	10.15	10.16	10.06	9.78	13.27

Gap closed by CB-VAR (%)

K	vs PL	vs LW
100	27.31	53.00
500	31.69	58.39
1000	30.03	22.36
2000	26.85	4.53

Monthly shrinkage intensity estimation errors, GMV

K	Mean error		MAE		MSE	
	PL	CB-VAR	PL	CB-VAR	PL	CB-VAR
100	0.01	0.04	0.27	0.24	0.12	0.08
500	−0.00	0.02	0.27	0.24	0.11	0.08
1000	−0.01	0.00	0.29	0.27	0.14	0.10
2000	−0.00	0.10	0.27	0.24	0.14	0.10

Appendix E — Nonlinear shrinkage (QIS): full tables

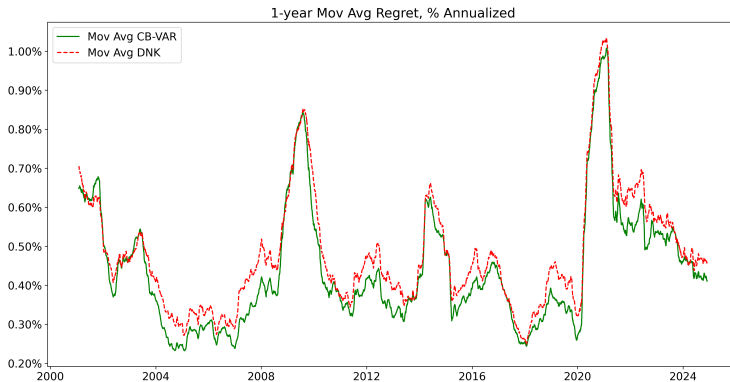
GMV realized annualized volatility (%)

K	QIS	QISCV	PL	CB-VAR	Oracle	Ridgeless	Gap closed vs PL
100	12.91	13.40	12.80	12.77	12.47	14.82	8.99%
500	10.99	11.21	11.02	10.92	10.78	13.49	39.58%
1000	10.01	10.16	10.17	10.08	9.83	11.99	26.21%
2000	10.10	10.29	10.40	10.28	9.94	13.27	25.83%

Long-only realized annualized volatility (%)

K	QIS	QISCV	PL	CB-VAR	Oracle
100	13.61	14.00	13.57	13.56	13.35
500	13.71	13.70	13.53	13.55	13.29
1000	13.02	12.97	12.98	13.00	12.66
2000	12.91	12.80	12.97	13.09	12.54

Appendix E — Nonlinear shrinkage: regret vs oracle ($K = 500$, 1Y moving average)



Appendix E — Nonlinear shrinkage: estimation errors

Monthly shrinkage intensity estimation errors, GMV

K	Mean error		MAE		MSE	
	PL	CB-VAR	PL	CB-VAR	PL	CB-VAR
100	0.04	0.09	0.49	0.51	0.47	0.30
500	0.01	-0.11	0.50	0.46	0.38	0.26
1000	0.00	-0.04	0.42	0.38	0.27	0.19
2000	-0.01	-0.10	0.34	0.35	0.19	0.16

- MAE can be higher while MSE is lower: CB-VAR avoids the few large errors that hurt the investor most.

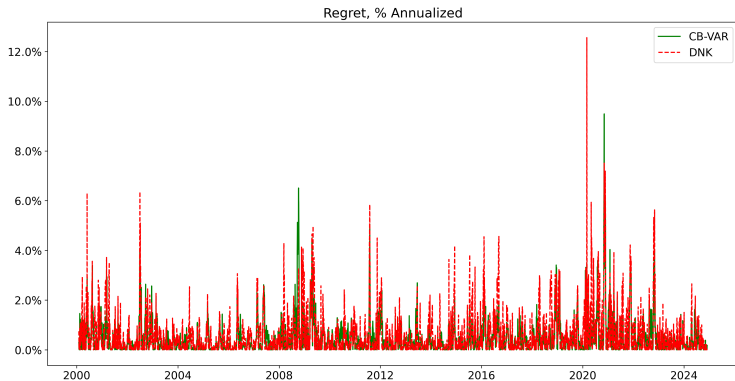
Appendix F — Statistical significance of the regret difference

Paired daily regret differences $d_t = r_{CB-VAR,t} - r_{PL,t}$; $H_0 : \mathbb{E}[d_t] = 0$.

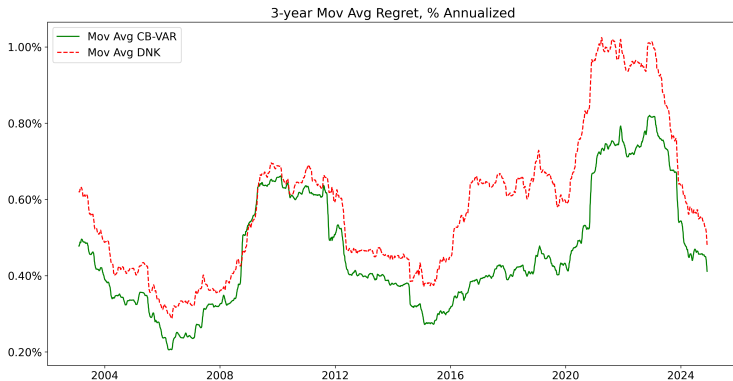
Linear shrinkage					Nonlinear shrinkage				
K	t-test	Wilcoxon	Perm.	99% CI	K	t-test	Wilcoxon	Perm.	99% CI
100	0.00**	0.00**	0.00**	[−0.13%, −0.08%]	100	0.00**	0.00**	0.00**	[−0.10%, −0.07%]
500	0.00**	0.00**	0.00**	[−0.14%, −0.11%]	500	0.00**	0.00**	0.00**	[−0.05%, −0.03%]
1000	0.00**	0.00**	0.00**	[−0.06%, −0.04%]	1000	0.00**	0.00**	0.00**	[−0.05%, −0.03%]
2000	0.00**	0.00**	0.00**	[−0.04%, −0.02%]	2000	0.00**	0.00**	0.00**	[−0.09%, −0.06%]

Summary statistics of daily regret, % ($K = 500$, linear)						
	Len	Avg	Std	25%	50%	75%
CB-VAR	6242	0.4543	0.6854	0.0525	0.2218	0.5873
PL	6242	0.5779	0.8353	0.0631	0.2686	0.7604

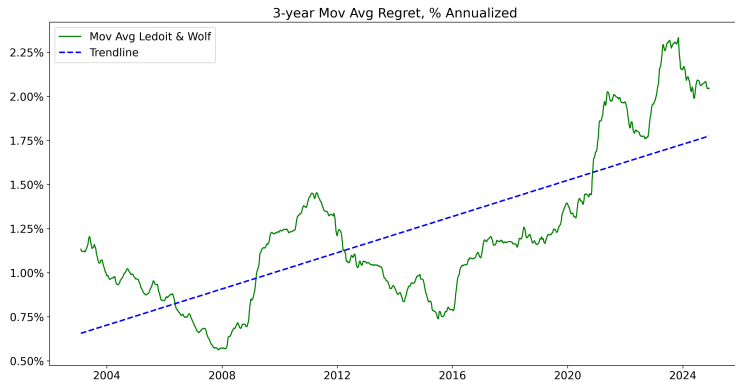
Appendix G — Daily 21-day-ahead regret ($K = 500$, linear)



Appendix G — Regret vs oracle, 3Y moving average ($K = 500$, linear)



Appendix G — Ledoit–Wolf regret, 3Y moving average ($K = 500$, linear)



Appendix H — Convexity of the linear-shrinkage reward curve

Let $\hat{\Sigma}(\lambda) = (1 - \lambda)\bar{\Sigma} + \lambda\bar{\sigma}^2 I_K$, with $A := \bar{\Sigma} - \bar{\sigma}^2 I_K$ and $B := \Sigma^* - \bar{\Sigma}$. Then

$$\|\Sigma^* - \hat{\Sigma}(\lambda)\|_F^2 = \|B + \lambda A\|_F^2 = \|B\|_F^2 + 2\lambda \langle A, B \rangle + \lambda^2 \|A\|_F^2,$$

a quadratic in λ with non-negative leading coefficient $\|A\|_F^2 \geq 0 \Rightarrow$ convex.

$\bar{\Sigma}$ — sample covariance; Σ^* — true covariance; $\|\cdot\|_F$ — Frobenius norm; $\langle \cdot, \cdot \rangle$ — Frobenius inner product.

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